

3.1.1 Kinematics

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) displacement, instantaneous speed, average speed, velocity and acceleration	<i>M0.1, M1.4, M3.7, M3.9</i> HSW10, 12
(b) graphical representations of displacement, speed, velocity and acceleration	<i>M3.6</i> HSW3 Using data-loggers to analyse motion.
(c) Displacement–time graphs; velocity is gradient	<i>M3.4, M3.7</i>
(d) Velocity–time graphs; acceleration is gradient; displacement is area under graph.	Learners will also be expected to estimate the area under non-linear graphs. <i>M3.5, M4.3</i>